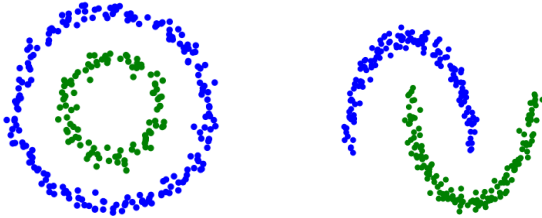


JuML Programming Project

Group A

Spectral clustering

- does a low-dimension embedding of the affinity matrix between samples, followed by KMeans in the low dimensional space.
- can detect non-convex patterns and linearly non-separable clusters.



- The module should implement some spectral clustering algorithms.
- Generate some test datasets.
- Build the affinity matrix.
- Perform the embedding of the data in the space spanned by the eigenvectors of the matrices derived from the affinity matrix.
- evaluate and visualize the results.
- compare clustering solutions.

related Julia and Python packages:

<https://github.com/lucianolorenti/SpectralClustering.jl>

<https://scikit-learn.org/stable/modules/generated/sklearn.cluster.SpectralClustering.html>

References:

- [1] [Normalized cuts and image segmentation](#), 2000 Jianbo Shi, Jitendra Malik
- [2] [A Tutorial on Spectral Clustering](#), 2007 Ulrike von Luxburg
- [3] [Multiclass spectral clustering](#), 2003 Stella X. Yu, Jianbo Shi
- [4] <https://papers.nips.cc/paper/2619-self-tuning-spectral-clustering.pdf>

Required milestones for session 1:

- Basic project structure available on GitHub (or equivalent)
- Some reviewable core functionality implemented, e.g. generation of test data and computation of the affinity matrix
- Minimal documentation, including a basic "Getting Started" guide

Due by Tuesday 09.06.2026